Rodrigo Fernando Ortiz Barahona

Computer Science Engineer

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Portfolio with video and screenshots of every system: explorador-perfiles-github.netlify.app • LinkedIn: linkedin.com/in/rodrigo-ortiz-b7a879232



PROFESSIONAL PROFILE

Computer Science Engineer with experience building business management systems (ERP) and SaaS applications with production automations from scratch. I built a custom ERP with inventory, production, catalog, roles and permissions and electronic-invoice import modules, and a multi-company platform with process automation: cron jobs, scheduled e-mails, OAuth integrations with external APIs and approval workflows. I complement this with AI/LLMs to automate tasks and I co-authored a peer-reviewed paper on agentic AI workflows (ICI2ST 2026). I am especially interested in business process automation and systems integration. Every project can be seen running at explorador-perfiles-github.netlify.app/en.

EXPERIENCE

Junior IT & Innovation Analyst - NovaCredit Banco Internacional | 2026

- Support the IT area on technology and innovation initiatives: process automation, evaluation of internal tools and technical documentation.

E-commerce Web Developer (Freelance) - Beltà Collective | Nov 2025 – Dec 2025 · beltàcollective.es

- Designed and launched an online store on WordPress: catalog, cart and payment gateway.
- Customized the interface with HTML, CSS and JavaScript/TypeScript, with on-page SEO and UX optimization.

IT Systems Assistant - MASOLUC (Chamber of Commerce) | Jun 2025 – Aug 2025

- Database migration, Active Directory implementation and update of the technology inventory.
- Technical support for equipment and reporting for the IT area.

Web Design Internship — Universidad del Azuay (IT Department) | Feb 2025 – May 2025

- Built landing pages and reusable templates with Tailwind CSS on Drupal.
- On-page/off-page SEO, site error diagnosis and fixing, and performance monitoring with web analytics tools.

SELECTED PROJECTS

ERP system for a jewelry workshop | Custom development · in production · 2026 · see it working

Python, FastAPI, PostgreSQL, SQLAlchemy/Alembic, Next.js, TypeScript, Docker, Nginx

- Complete ERP: inventory (kardex, movements, stock, archiving), production (configurable processes and stages, orders, quality control, assembly recipes), catalog, product types, units and a modular REST API.
- Automated inventory entries by importing and parsing XML electronic invoices, with supplier code aliases and traceability to the original document.
- Internal request flow between production and inventory with states and in-app notifications, and automatic transfer of finished production to inventory.
- Role-based access control with granular permissions, JWT in HttpOnly cookies, brute-force rate limiting and CSRF/CORS protection.

- Production deployment with Docker Compose, Nginx, automatic migrations on startup and a pg_dump backup strategy.

TaskKeep - Multi-company business task manager (SaaS) | In production · task-keep-lime.vercel.app · see it working

Next.js, TypeScript, Supabase (PostgreSQL), Resend, Google Drive API, Docker, Vercel

- Multi-tenant platform with 3 roles (admin, manager, collaborator), task and file approval workflows, calendar, action audit log and custom JWT authentication.
- Automated e-mail reminders: daily cron scheduling recurring sends (daily, monthly, N days before the deadline) through the Resend API, handling time zones and rescheduling automatically when tasks are edited.
- Google Drive API integration (OAuth 2.0 per company): automatic folder creation per task, validated file uploads and an approval flow that moves documents.
- Cross-cutting security: rate limiting, origin validation, encrypted tokens and continuous deployment with Docker and Vercel.

Molaris - Dental clinic management system (SaaS) | Custom development · 2026 · see it working

NestJS, Prisma, PostgreSQL, Next.js, TypeScript, Tailwind CSS, Turborepo, Docker, Caddy, Jest/Vitest

- Complete clinical record: medical history, visits, ICD-10 coded diagnosis, interactive odontogram with Ecuador's form 033 symbols, procedures, certificates, prescriptions, quotes and scheduling.
- Multi-tenant version with one database per clinic and a central control plane; architecture decisions documented as ADRs and automated quality checks before every push.

Verbum - AI Bible mobile app for older adults | Personal project · see it working

React Native, Expo, TypeScript, Supabase, SQLite, generative AI

- Full offline Bible (SQLite), topic search and a daily AI-generated video (passage, narration and scene) in a reels-style feed, with an accessible large-type interface.

Primero en Leonida - Subscription giveaway platform | Personal project · see it working

Next.js, Supabase (PostgreSQL, RLS), Lemon Squeezy, Tailwind CSS, Vitest

- Verifiable monthly draws (public seed and full published list) and a community achievements game by country; the rules that decide a winner live in the database, with anti-cheat tests.

El Botón Rojo - Real-time competitive web game | Personal project · see it working

Node.js, Fastify, WebSockets, Redis, PostgreSQL, React, Vite, Caddy

- Live top-10 leaderboard, batched clicks and layered anti-cheat (token bucket, timing plausibility, regularity scoring and human verification) with an auditable winner per round.

PlantChain - Blockchain tree sponsorship web platform | Personal project · Hult Prize 2026 · see project

- Token purchase system with smart contracts; each token immutably records the tree's planting and history.
- Automated recording of events (planting, care, growth) and geolocated tree visualization.
- Startup presented at the Hult Prize at Universidad del Azuay (February 2026).
- **Other projects:** installable PWA with push notifications (Vercel Functions, Vercel Blob, OneSignal) and a GitHub profile explorer (NestJS + Next.js), the base of my portfolio.

PUBLICATIONS & AWARDS

- **Peer-reviewed paper:** Erazo-Garzón L., Chimbo J., Ortiz R., Illescas L., Cedillo P. “A Domain-Specific Language for Modeling and Implementing Agentic AI Workflows”. Presented at ICI2ST 2026 (EPN, Quito, Sept. 2026); to appear in MDPI Engineering Proceedings.
- **Certificate:** Hult Prize at Universidad del Azuay (February 2026), as a startup team member of PlantChain.

EDUCATION

B.Eng. in Computer Science — Universidad del Azuay, Cuenca | 2022 – 2026

High School Diploma in Sciences — U.E. San José de Calasanz, Cañar | 2017 – 2022

TECHNICAL SKILLS

- **ERP/CRM systems:** design and development of inventory, production, catalog, electronic invoicing (XML) and role/permission modules.
- **Automation & integrations:** cron jobs, scheduled e-mail (Resend), OAuth 2.0 and third-party APIs (Google Drive, payment gateways), webhooks and REST APIs.
- **Backend:** Python (FastAPI), Node.js (NestJS, Fastify), PHP, Java, C++; microservices and REST APIs; PostgreSQL, MySQL, Redis and NoSQL databases.
- **Frontend & mobile:** JavaScript/TypeScript, React, Next.js, React Native (Expo), Tailwind CSS, HTML/CSS; accessibility, performance and SEO.
- **DevOps:** Docker, Nginx, Caddy, Git/GitHub, deployment on Vercel, Netlify, Render and own servers; database backups and migrations.
- **Applied AI:** integrating LLM APIs into applications, building assistants, prompt engineering and automation with AI tools.

LANGUAGES

Spanish: native **English:** B1 intermediate